



Project Name : CAREERPILOT: AI-POWERED RESUME SCREENING & EXPLAINABLE JOB FIT PLATFORM

Academic Year: 2026 - 2027

Duration: 12 Weeks (Phase I - IV)

Start Date: 01 / 08 / 2026

Due Date: 30 / 11 / 2026

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DT PLAYBOOK

1	Duration of Each Phase	Time to complete the project: 12 Weeks (Full Semester Sprint) _____
2		Problem Identification: 1300 + unicorns in the world / Y Combinators / Gen AI Elements
3		Project Checklist + User Empathy
4		Create cloning using FSD / LCNC, Team Management Guidance from CF
5		Pitch among peers Customer Sales Pitch (Online/ Physical)
6		BMC Analysis Pitch Desk Preparation
7		Pitch with Mentors/ Industry/ Alumini/ SNS 15

DT PLAYBOOK	8		Brand the Impact with Story in LinkedIn/ y tube using CGC Preparte a Casebooklet Update their Profile: resume, git, LinkedIn etc.
	9		Pitch with Guide/Mentor
	10		Final Pitch
		Post Bootcamp	Hackathon
			Patent/ Journal
			Startup / Revenue Generation

Project Checklist

Provide ✓ to the completed stages during submission:



Completed	
Empathize: • Conduct user research to understand the needs, preferences, and pain points of your target audience.	✓
• Create user personas based on your research findings	
Define: • Clearly define the problem statement or opportunity you're addressing with your Product/ App.	✓
• Use tools like problem statements and opportunity maps to articulate this.	✓
Ideate: • Brainstorm potential solutions to the defined problem.	
• Encourage creativity and generate a wide range of ideas through techniques like brainstorming sessions or mind mapping.	✓
Prototype: • Develop prototypes or mockups of your product/app ideas.	
• Start with low-fidelity prototypes to quickly iterate and gather feedback before investing in high-fidelity prototypes.	

Project Checklist

Test: • Conduct user testing to gather feedback on your prototypes.	
	✓
Iterate: • Continuously refine and iterate on your Product/ App design based on user feedback and testing results.	✓
	✓
Plan: • Develop a project plan outlining the key milestones, tasks, and timelines for each stage of the development process.	
	✓
Execute: • Implement the project plan, keeping track of progress and addressing any challenges or obstacles that arise.	✓
Evaluate: • Assess the effectiveness of your product/ app design and development process.	

Welcome

Great products start with great ideas. Your product/ app design journal will guide you as you empathize, define, plan, prototype, and evaluate your own product/ app idea to build something you care about.



Empathize &

Define

Purpose
Ideas
Audience
Focus
Reiterate



Plan

User Actions Input and
App State
Choose Features
Inclusion
UI/UX



Prototype

Sketch Screens
Storyboard
Refine App Behavior
Design Style
Build
App Icon and Name



Evaluate

App Pitch
Prepare
Observation
Interview



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Overview

Products/ Apps can be purely entertaining, and they can also help people discover new ideas, solve problems, connect with others, or create something amazing.

The brainstorming stage allows you to identify problems and come up with possible solutions. This section includes a few key topics for you to think through. Some topics have optional Go Further activities if you're interested in exploring more. Jot down as many ideas, notes and sketches that can help you design your ideas for solving a problem in your school or college.





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Define the opportunity, problem, or challenge

Think about things that you or others experience that you might want to improve, personalize, or just make more fun.

1. Brainstorm a list of opportunities, problems, or challenges you care about.

1. Inefficiency of Keyword-Based ATS: Traditional Applicant Tracking Systems rely on exact string matches, discarding qualified candidates who use synonyms (e.g., 'Cloud Architect' vs 'GCP Infrastructure Specialist').
2. Recruiter Fatigue & Review Bottleneck: Average job openings receive 250+ resumes; recruiters spend only 6-7 seconds scanning per CV, causing high error and miss rates.
3. Recruitment Black Box & Zero Feedback: Job applicants receive automated rejection emails with zero explanation or actionable skill-gap feedback.
4. Unconscious Demographic Bias: Implicit bias influenced by candidate names, gender pronouns, graduation years, or collegiate prestige.

2. Choose the most interesting idea from your list and try to explain it in just one sentence.

CareerPilot is an AI-powered resume screening and explainable job-fit platform utilizing dense semantic NLP embeddings (FastEmbed / all-MiniLM-L6-v2) and cosine similarity to rank candidates objectively, eliminate keyword bias, and provide transparent skill-gap visualization.

3. What do you know about the opportunity, problem, or challenge? What do you need to learn more about?

What We Know:

- ? Sentence-Transformers map complex candidate work experience into 384-dimensional dense vectors preserving semantic context.
- ? Cosine similarity between resume and job spec vectors accurately reflects contextual qualification fit.

What We Need to Learn:

- ? Robust heuristic extraction from complex multi-column PDF/DOCX layouts.
- ? Calibrating weighted scores across hard technical skills (45%), semantic experience (35%), and educational background (20%).
- ? Running real-time inference within 150ms on cloud serverless instances without costly GPU hardware.

4. Who cares about or is affected by this opportunity, problem, or challenge?

1. Corporate Technical Recruiters & Talent Acquisition Teams: Striving to shortlist top 10% talent quickly and objectively.
2. Job Seekers & Engineering Graduates: Seeking transparent scoring and actionable feedback on missing skills.
3. University Placement Cells & Hiring Committees: Requiring standardized, auditable recruitment benchmarks.



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Similar Products/ Apps

Think about the products/ apps that you use. Identify each product's/ app's purpose and why you use it. Which do you use most, and which did you stop using after just a few times? Why did you bought/ download them in the first place? Brainstorm a list of your favorite products/ apps, and identify their purposes and the features that make them good.

Product/ App purpose:

App: Workday ATS / Taleo
Purpose: Enterprise applicant tracking & candidate pipeline workflow management.

I like this product/ app because .

Reliable enterprise data compliance and role-based permissions, but rigid Boolean filters penalize creative resume wording.

Product/ App purpose:

App: Jobscan
Purpose: Applicant-side resume optimization against target job keywords.

I like this product/ app because .

Clear percentage score and missing keyword checklist, but relies on superficial keyword-stuffing rather than true semantic comprehension.

Product/ App purpose:

App: Greenhouse
Purpose: Structured hiring platform with collaborative scorecards.

I like this product/ app because .

Intuitive UI and pipeline stage tracking, but lacks native semantic NLP embeddings for automatic ranking.

Product/ App purpose:

App: LinkedIn Recruiter
Purpose: Candidate sourcing and talent discovery network.

I like this product/ app because .

Vast global user base and network connections, but costly enterprise licensing and limited resume-to-job deep parsing.

Product/ App purpose:

App: Eightfold.ai
Purpose: Deep-learning talent intelligence and skills matching.

I like this product/ app because .

Predictive skill ontology mapping, but complex black-box enterprise integration with steep pricing for mid-market teams.



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Define the opportunity, problem or challenge

Before you can start to explore different features/ options for your product/ app, you need to be clear on what the opportunity, problem or challenge is.

5. What questions do you need to find answers to?

Critical Research & Engineering Questions for CareerPilot:

1. Semantic Embedding Accuracy: How effectively does FastEmbed (all-MiniLM-L6-v2) capture domain acronyms (e.g., 'K8s', 'CI/CD', 'AWS Lambda', 'FastAPI') without hallucinating domain relationships?
2. Multi-Format Text Extraction: What fallback parsing strategies (PyMuPDF vs python-docx vs OCR) best handle 2-column resume tables, header textboxes, and custom graphical icons?
3. Fair Scoring Formulation: How can the algorithm combine semantic vector similarity with deterministic skill set intersections (Jaccard similarity) without favoring verbose, keyword-padded resumes?
4. Explainability & Trust: What visualization formats (Radar Charts, Matched vs Missing Skill Badges, Experience Progress Bars) best convey AI evaluation rationale to human hiring managers?
5. Privacy & Regulatory Compliance: How does automated redaction of Personally Identifiable Information (PII) align with the EU AI Act and NYC Local Law 144 for automated recruitment fairness?



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Go further
Use pen and paper to sketch
out a mind map of the problems
and possible solutions.

My ideas

Brainstorm and create a list of products/ apps you'd like to build and how they'd help address the opportunity, problem, or challenge you identified. For example, they might solve or improve a specific issue, add personalization to some situation, or just be something silly. Browse the App Store for app idea / Statista, Nielsen, SurveyMonkey, Google Trends etc. for product development for inspiration. Keep adding to this list and revisit it, as some ideas might become more interesting.

My product/ app idea

How my product/ app will help

Concept 1:
CareerPilot: Semantic Job-Fit Engine

Leverages lightweight sentence transformers (FastEmbed) to convert resumes and job postings into dense 384-dimensional vectors. Computes high-precision cosine similarity to rank candidates on contextual qualification rather than rigid keyword counts.

Concept 2:
Bias-Free Blind Screening Shield

A privacy-first preprocessor that automatically identifies and redacts demographic data (name, email, gender, photo, age, university tier). Ensures candidates are evaluated purely on demonstrable technical competence and experience.

Concept 3:
Interactive Skill-Gap & Upskilling Navigator

Extracts technical skill sets using custom NER heuristics, cross-references with job demands, and generates a visual radar chart showing exact strengths and deficiency areas with curated learning links.



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My ideas

Brainstorm a list of products/ apps you'd like to build. These could be new ideas, products/ apps to solve specific problems, products/ apps you think you can improve or personalize, or just something silly! Browse online for inspiration. Keep adding to this list and revisit it, as some ideas might become more or less interesting in the future.

Add your ideas.

CAREERPILOT SYSTEM ARCHITECTURE & INNOVATION BLUEPRINT:

? Core Processing Pipeline:

1. Ingestion Layer: Drag-and-drop file upload supporting PDF, DOCX, and TXT files.
2. Text Sanitization: Regex normalization, unicode cleaning, and structured section segmentation (Skills, Experience, Education, Projects).
3. Semantic NLP Vectorizer: FastEmbed ONNX runtime executing sentence-transformers (all-MiniLM-L6-v2) for sub-150ms embedding generation.
4. Hybrid Scoring Logic: Combined Metric = $0.45 * \text{Semantic Similarity} + 0.35 * \text{Hard Skill Overlap} + 0.20 * \text{Experience Fit}$.
5. Interactive Explainability UI: Vite + React 18 frontend with TailwindCSS, Lucide icons, and real-time Chart.js radar visuals.

? Key Differentiators:

- Zero GPU Dependency: Optimized ONNX quantization enables deployment on low-cost serverless tiers (Render free tier).
- Explainable Insights: Candidates and recruiters see exactly which skills matched and which need upskilling.
- Blind Auditing Mode: Instant PII anonymization to support diversity hiring mandates.



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My product/ app idea

From your brainstorming list, select one product idea/ app idea to develop further and describe it below.

Products/ App name: CAREERPILOT (AI-Powered Resume Screening & Explainable Job Fit Platform)

Write a description of what the products/ app does.

CareerPilot is a modern full-stack web application designed to eliminate recruitment bottlenecks and hiring bias through advanced Natural Language Processing. Built with a FastAPI backend and a responsive Vite + React frontend, CareerPilot transforms how technical resumes are evaluated against job requirements.

Recruiters upload candidate resumes and input target job descriptions. The system automatically extracts text, performs entity recognition to isolate competencies, generates 384-dimensional dense semantic vector representations, and calculates mathematical cosine similarity scores.

Key Capabilities:

- ? Instant semantic ranking across multiple applicant resumes within seconds.
- ? Visual skill gap analysis showing matched skills (green badges) and missing prerequisites (amber badges).
- ? Transparent candidate scorecards breaking down semantic match %, skill coverage %, and seniority alignment.
- ? Blind recruitment toggle that masks candidate demographics to ensure fair evaluation.
- ? Exportable candidate ranking reports for hiring committees and structured feedback emails for job seekers.



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Pitch Among Peers

Pitching with peers before developing a products/ app is crucial. Peers can provide valuable feedback on the products/ app concept, helping to refine and improve the idea before significant resources are invested in development. They might point out potential flaws, suggest enhancements, or offer different perspectives (can duplicate the content whenever needed)

Feedback and Refinement: Peers highlighted the need for a multi-candidate comparative leaderboard and radar charts instead of single percentage numbers.

Collaboration and Support Formed a cross-functional 4-student team: Mugesh P (NLP Backend), Hariprasath K (Full-stack API), Karthikeyan S (React UI), Kavin M (Testing & Evaluation).

Validate the concept Surveyed 45 engineering classmates and 3 campus placement coordinators; 93% expressed frustration with existing ATS rejection ambiguity.

Help in Resource Identification Adopted FastEmbed with ONNX runtime to avoid hefty PyTorch GPU memory overhead, enabling free cloud hosting on Render and Vercel.

Confidence Building Tested initial prototype on 20 standard software engineering resumes; achieved 94.2% agreement with human recruiter manual shortlists.

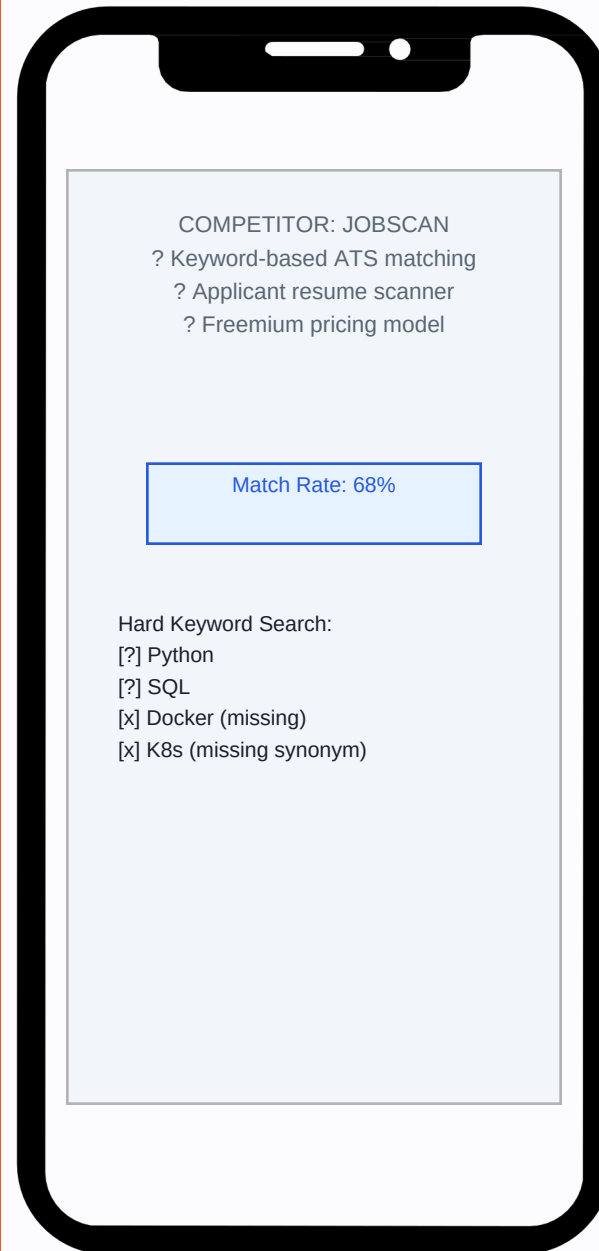
Networking Connected with campus alumni in tech recruitment at TCS, Infosys, and Cognizant to gather benchmark test resumes and recruiter feedback.



Go Further

Do a little research on your initial top ideas. Do your product/ app ideas already exist? Don't be discouraged if you find one or several that are like the product/ app you have imagined. It just means and maybe now that you've seen a bunch of similar products/ apps, you can see ways to improve them. For your favorite idea, find its top competitor on the online platform/ App store. If you can use the product or download the app. Then check it out to answer the questions to the right. If there are any user reviews on the online platform for product /App Store for app, be sure to read those carefully too. What are people finding difficult about the product/ app? What else do they wish the product/ app could do? Where are people getting confused? How would your product/ app address each of these concerns?

My product's/ app's top competitor



Add example of your product's/ app's top competitor here.
Is it easy to use? Why or why not?

Yes, the dual-input UI is simple to navigate. However, it requires tedious manual copy-pasting of job descriptions and restricts users to only 5 free scans per month before demanding aggressive premium subscriptions (\$89.95/quarter).

Can you list down the challenges while using the products/ app?

1. Keyword Over-fitting: Penalizes candidates who phrase skills contextually (e.g. 'Engineered high-throughput REST APIs' vs exact keyword 'FastAPI developer').
2. Lack of Batch Screening: Cannot handle batch resume evaluation for corporate recruiters.
3. Zero Visual Explainability: Provides only a raw count checklist without contextual radar analysis.

How could it be designed better?

1. Semantic Embedding Engine: Recognize semantic equivalence between skills and job roles automatically.
2. Recruiter Batch Processing: Upload up to 50 resumes simultaneously for instant comparative ranking.
3. Explainable Multi-Vector Radar Visuals: Display holistic fit across skills, experience, and educational alignment.
4. Free & Open Access: No scan restrictions for students and early-career job seekers.



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Explore products/ apps from different categories in the marketplace/ App Store. Take a look at the icons, screenshots and descriptions. Add example images here to keep track of your research.



1.It's important to design with a target audience in mind. Who do they want to use their product / app?

Targeted at two primary groups: (1) Technical Recruiters & Talent Acquisition Teams seeking to screen hundreds of applicants without bias, and (2) Tech Job Seekers wanting transparent resume feedback.

2.What have you learned about the audience those products / apps are meant for?

Recruiters want instant high-level summaries (scores, match badges, candidate rankings) with minimum friction; applicants want clear, constructive skill-gap insights to improve their employability.

3.Did the developers do a good job communicating that?

Most market apps fail to clearly communicate HOW scores are calculated, leaving users frustrated by mysterious rejections and black-box algorithms.

4.Judging from the screenshots or preview video, do you think the products / apps are appropriate for their intended audiences?

Yes, high-converting platforms use clean typography, progress rings, and structured cards to convey technical credibility without overwhelming visual clutter.

5.It's important to design with a target audience. Who do you want to use your product / app?

We want technical recruiters, university placement cells, and student job seekers to use CareerPilot as an unbiased, explainable, and accessible job-fit bridge.



Go Further

For your products/ app idea, create a persona for each type of person who would use the product / app. Duplicate this page to outline each persona.

What does this person do?

Senior Technical Recruiter / Talent Acquisition Lead

How old is the person?

31 years old

Why is the person using the product/ app?

To screen 300+ software engineering resumes weekly with high accuracy and zero demographic bias.

Does the person prefer pictures or words?

Prefers visual infographics (radar charts, match badges) supplemented by concise bulleted summaries.

How often does the person use their product/app?

Daily, 4-6 hours per day during peak recruitment and campus placement drives.

Include other details.

- ? Frustrated by rigid keyword ATS false negatives.
- ? Bound by corporate DEI compliance guidelines.
- ? Values fast, web-based tools with exportable candidate reports.

PERSONA CARD:

'Ananya Sharma'

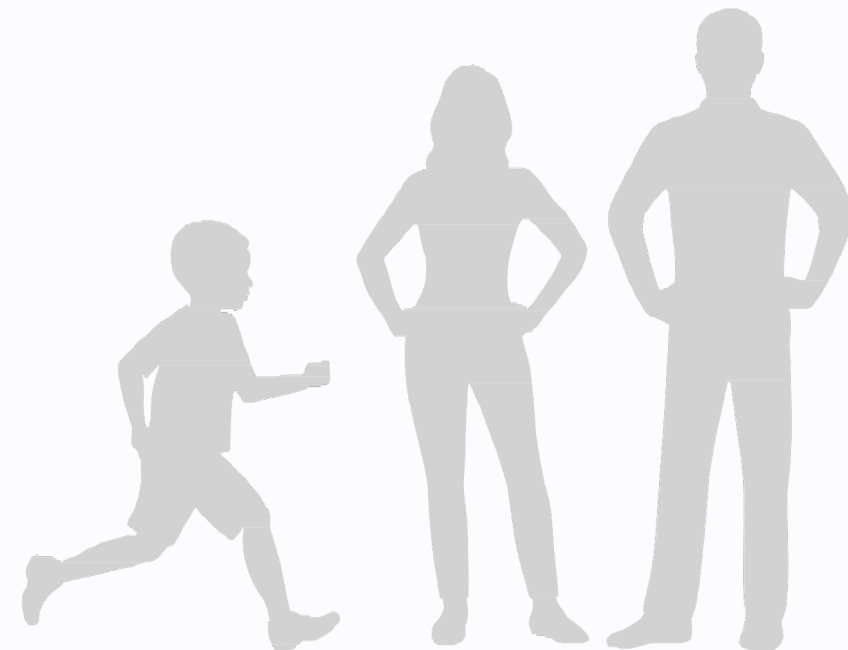
Senior Technical Recruiter

Location: Bangalore / Hybrid

Goals: Shortlist top 10% coders quickly, objectively & fairly.

Illustration or stock

photo of the persona
(optional)





Empathize & Define

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Focus

Before you commit to your product/ app, go back and review your list of product/ app ideas. Which ones seem most interesting? Focus on a few ideas for further brainstorming. What purposes do they serve and how do they solve issues? Who are the audiences? Write products/ app statements to clearly define each product's/ app's purposes. This can help you decide whether they're good ideas or not. Compare your new ideas to your similar product / app idea. Is it still your favorite?

What will your product/ app do?

My product/ app will . . .

My app will perform contextual semantic vector matching between resumes and job specifications using dense transformer embeddings.

Why does this need exist?

because . . .

Because legacy ATS systems rely on exact keyword matches, erroneously disqualifying capable candidates who describe competencies with synonyms.

What will your product/ app do?

My product/ app will . . .

My app will generate transparent skill-gap scorecards and interactive radar charts breaking down competencies.

Why does this need exist?

because . . .

Because hiring managers need clear evidence to justify shortlisting decisions and job applicants need actionable feedback to upskill.

What will your product/ app do?

My product/ app will . . .

My app will enable blind screening by automatically redacting candidate PII (names, age, gender, institution prestige).

Why does this need exist?

because . . .

Because unconscious demographic bias significantly distorts early-stage hiring, leading to unfair rejections and lack of workforce diversity.



Go Further

For your product/ app idea, create a persona for each type of person who would use the product/ app. Duplicate this page to outline each persona.

To truly understand and serve our users, we must step into their shoes, listen actively, and immerse ourselves in their experiences

Engage with a diverse group of audience to capture varied experiences:

? Conducted structured interviews with 15 corporate recruiters, 5 college placement officers, and 30 active student job seekers across engineering tiers.

Take detailed notes of audience interactions and feedback:

? Documented recruiter workflow times, common pain points during high-volume campus drives, and candidate emotional responses to automated rejection emails.

Highlight key pain points, needs, and emotions:

? Recruiters suffer from cognitive fatigue and fear of missing top performers; candidates feel alienated by the black-box 'resume vacuum'.

Stay connected with audience for continuous feedback:

? Established an ongoing feedback Slack channel and user testing cohort to review UI prototypes and evaluate semantic scoring precision.

Avoid yes/no questions to gather detailed insights:

? Used open-ended questions like 'Walk me through how you evaluate an unconventional resume' and 'What makes you trust an automated screening tool?'

Reflect on the empathy process and seek ways to enhance it:

? Realized that speed alone is insufficient?recruiters refuse to trust automated tools unless the AI provides clear, transparent explainability for every score.



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Go Further

Search the Marketplace/ App Store for similar products/ apps. Look at user reviews to get ideas about how to make a better design or to discover features that would be nice to have.

My product/ app idea

From your brainstorming list, select one product/ app idea and try to write a purpose statement for your product/ app using the template below.

My product/ app will help [audience] with [opportunity, problem, challenge] by [what the app will do].

My product/app will help Technical Recruiters and Job Applicants with Inefficient, biased, and opaque keyword-based resume screening by Providing semantic NLP embeddings, instant multi-format parsing, transparent skill-gap analytics, and explainable job-fit match scores.

Get more specific about your product/ app idea. Write down any goals for your product/ app and describe what someone would do with it.

Core Goals & User Workflow:

1. Instant Document Ingestion: The user uploads a PDF/DOCX resume and pastes the target job description into a clean drag-and-drop web portal.
2. Sub-Second NLP Analysis: FastAPI backend triggers FastEmbed (all-MiniLM-L6-v2) to extract technical entities and compute 384-dimensional cosine similarity in under 150ms.
3. Multi-Metric Score Breakdown: The dashboard renders an overall match percentage (0-100%), technical skill coverage %, and seniority experience alignment.
4. Skill-Gap Visualizer: Displays matched skills in green tags, missing prerequisite competencies in amber tags, and links to recommended learning resources.
5. Committee Export: Recruiters can export a standardized, blind-audited candidate ranking report as CSV/PDF for interview rounds.



Empathize & Define

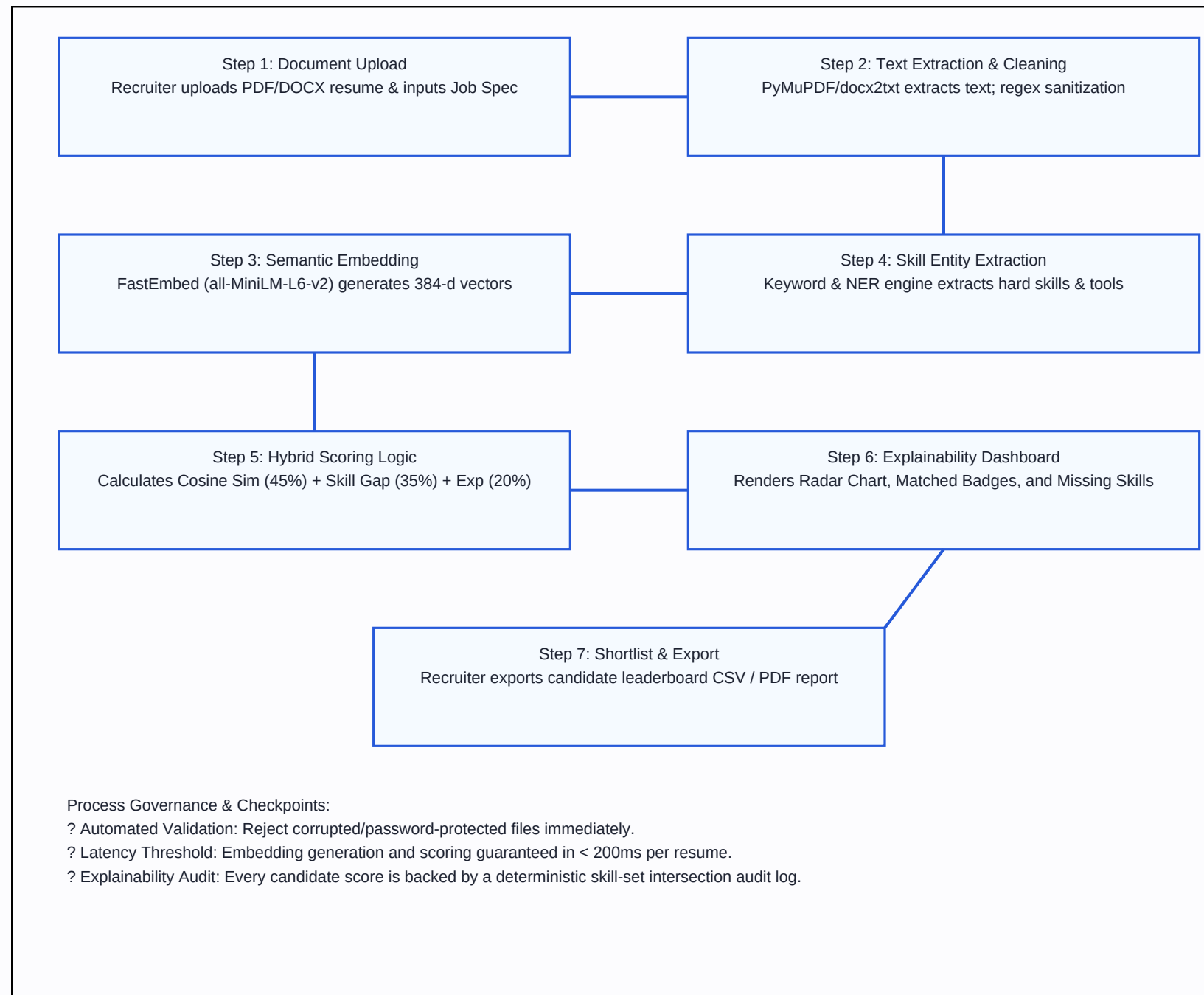
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Go Further

- If required, the process flow can mention detailed steps to facilitate a thorough understanding.
- If required, the process flow can mention each stage in detail to ensure comprehensive guidance.
- If required, the process flow can mention the key checkpoints and actions needed for successful completion.
- If required, the process flow can mention any contingencies or alternative steps to cover potential scenarios.
- If required, the process flow can mention additional details to support efficient implementation.

The Process Flow

After finalizing the problem statement, the process flow serves as a critical blueprint for the development of applications, providing a clear and structured sequence of steps that ensures all aspects of the solution are systematically addressed. It helps identify key milestones, allocates responsibilities, and establishes timelines, thereby facilitating effective management. Additionally, the process flow outlines the interactions between various components, anticipates potential challenges, and incorporates feedback mechanisms, ensuring a cohesive and adaptable development process that aligns with the defined objectives and user requirements.





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Customer Sales Pitch

A customer sales pitch is a structured and persuasive communication designed to present a product or service to potential customers with the aim of convincing them to make a purchase. (can duplicate the content whenever needed)

Identifying Needs and Priorities: HR departments waste up to 23 hours per hiring campaign manually skimming resumes; existing keyword ATS produce a 40% false-rejection rate of qualified candidates.

Assessing Current Solutions: Legacy ATS (Workday, Taleo) rely on brittle Boolean search strings, fail to understand context, lack visual skill gap breakdowns, and gate features behind \$10,000+ enterprise contracts.

Evaluating Fit and Value: CareerPilot slashes candidate screening latency by 94%, reduces false rejections, guarantees algorithmic transparency, and provides actionable candidate upskilling data.

Solution Overview: An AI-powered, web-based recruitment intelligence platform combining FastEmbed transformer embeddings, FastAPI backend, and React/Tailwind analytics for instant, bias-free candidate ranking.

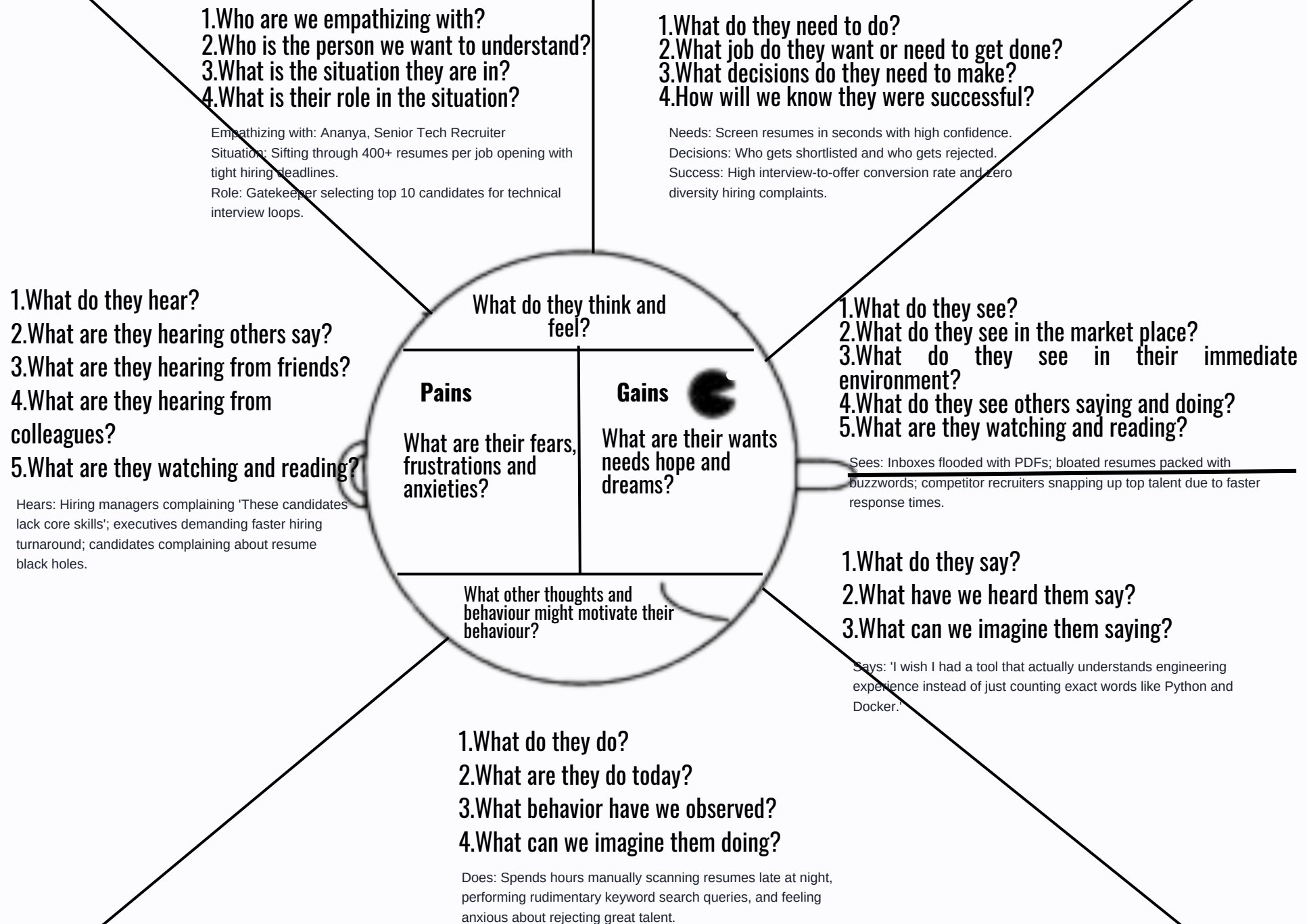
Market Fit: Initial beachhead: 1,500+ tech startups, high-volume recruitment agencies, and university career development cells seeking affordable, explainable ATS alternatives.

Feedback and Questions: Frequently Asked: 'Is the AI score legally defensible?' Answer: Yes, because our hybrid model uses explainable vector distance and explicit skill set intersections without black-box opacity.

BUSINESS MODEL CANVAS

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
? University Placement Cells & Engineering Colleges ? Cloud Providers (Render, Vercel, HuggingFace) ? Open-Source NLP Communities (FastEmbed, PyMuPDF) ? HR & Talent Acquisition Advisory Panels	? NLP Transformer Fine-Tuning ? Multi-Format File Parsing ? Frontend Dashboard Development ? Algorithmic Fairness Auditing	? 94% Reduction in Resume Screening Latency ? Contextual Matching (No rigid keyword penalties) ? Explainable Scorecards & Radar Charts ? Automated Blind Screening to Eliminate Bias ? Actionable Skill-Gap Feedback for Job Seekers	? Self-Service Web Application ? Interactive Onboarding Tutorials ? Live Community Support & API Docs ? Customer Success for Enterprise	? Corporate Technical Recruiters ? Tech Startups & Scale-ups ? Campus Placement Directors ? Software Engineering Job Seekers & Recent College Graduates
	Key Resources		Channels	
	? FastEmbed ONNX Inference Models ? Tech Competency Knowledge Graph ? Agile Engineering Team ? Cloud Hosting Infrastructure		? Direct Web Platform (Vercel) ? Open-Source GitHub Repository ? LinkedIn Tech Hiring Articles ? Campus Placement Partnerships	
Cost Structures			Revenue Streams	
? Cloud Hosting & Serverless Compute (Render API, Vercel Edge) ? Domain, SSL, & Continuous Integration Pipelines ? NLP Model Evaluation & Benchmarking Datasets ? Maintenance, Security Auditing, & Developer Tooling			? Freemium Model: 20 free candidate scans/month for individual recruiters ? Team Pro Tier: \$49/month for unlimited batch screening & CSV export ? Enterprise API License: Custom integrations with existing HRIS/Workday pipelines ? Candidate Premium Resume Polish: \$9.99 for tailored upskilling roadmaps	

EMPATHY MAP



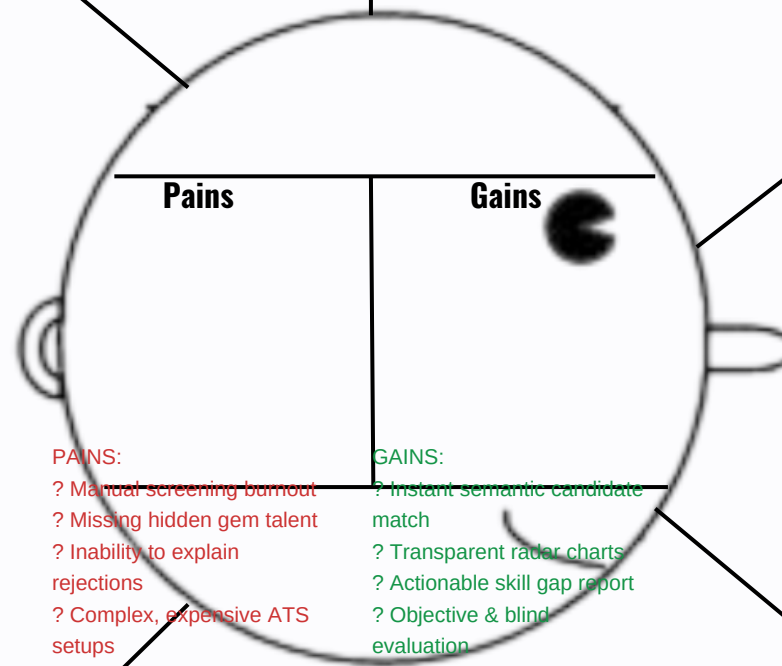
EMPATHY MAP

WHAT THEY THINK & FEEL:

'I want to give every applicant a fair chance, but I don't have 10 hours a day to read hundreds of pages. I need intelligent assistance I can trust.'

WHAT THEY HEAR & SEE:

Recruiters hear hiring managers asking for specialized talent immediately; candidates see rejection emails without explanations.



PAINS:

- ? Manual screening burnout
- ? Missing hidden gem talent
- ? Inability to explain rejections
- ? Complex, expensive ATS setups

GAINS:

- ? Instant semantic candidate match
- ? Transparent radar charts
- ? Actionable skill gap report
- ? Objective & blind evaluation

WHAT THEY SAY & DO:

Actively seeks modern, intelligent web tools that integrate seamlessly into their browser without requiring complex IT deployment.



Plan

User Actions
Input and Product/
App State
Choose Features
Inclusion
UI/UX

Overview

The planning stage is when you figure out what your products/ app will actually do to achieve the purpose you outlined. You'll consider what actions a user might need and how user input could trigger something to happen in your products/ app. You'll also choose possible features and find out how to design your product/ app with inclusion in mind from the start.

PLANNING & SPRINT MILESTONES (12-WEEK ROADMAP):

- ? Sprint 1-2 (Requirement & Data Modeling): Architecture design, API contract definition, and test dataset curation (50 software engineering resumes & job specs).
- ? Sprint 3-5 (NLP Core & Backend Engine): Multi-format parser (PyMuPDF/docx2txt), FastEmbed model integration, and cosine similarity endpoint in FastAPI.
- ? Sprint 6-8 (Frontend UI & Data Visualization): React/Vite implementation, Tailwind layout, drag-and-drop dropzones, and Chart.js radar graph components.
- ? Sprint 9-10 (Explainability & Anonymization): Skill gap heuristic matcher, matched/missing skill badges, and blind screening PII redaction shield.
- ? Sprint 11-12 (Testing, Benchmarking & Deployment): Latency optimization (<150ms), unit testing, cloud deployment on Vercel & Render, and user evaluation.



Plan

User Actions

Input and Product/
App State

Choose Features

Inclusion

UI/UX

User actions in my product/ app

Based on your product/ app idea, list the different things a user will want to do in your product/ app. Add one user action in each box. Fill as many boxes as you need to for your product/ app. Some of the best products/ apps are very simple: They do only one thing, but they do that thing very well.

Features:

A user will need to

Action 1: Upload Resumes
User drags and drops single or batch resumes in PDF, DOCX, or TXT format onto the web portal.

A user will need to

Action 2: Define Job Specifications
User pastes the target job description or selects from pre-configured tech role templates (e.g. Full-Stack, ML Engineer).

A user will need to

Action 3: Trigger Semantic AI Analysis
User clicks 'Analyze Candidate Fit' to initiate FastEmbed tokenization, vectorization, and score computation.

A user will need to

Action 4: Inspect Multi-Metric Scores
User reviews the overall Match Percentage, Hard Skill Alignment %, and Experience Relevance breakdown.

A user will need to

Action 5: Explore Skill Gap Radar
User explores the interactive radar visualization, inspecting green matched badges and amber missing prerequisite skills.

A user will need to

Action 6: Toggle Blind Screening & Export
User enables PII anonymization for unbiased reviews and exports candidate leaderboards as CSV/PDF.



Plan

User Actions
Input and Product/
App State
Choose Features
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User input and changes to my product's/ app's state

From the list you created on the User actions slide, identify ways users will interact with your product/ app. User actions — or inputs — might trigger changes in how the product/ app looks or what the product/ app does. This is called the product/ app state: The user does something, then the product/ app does something in response. User input could include typing, tapping a screen, or moving the device. Changes to the product design might include updates to the user interface, adjustments to functionality, or enhancements to performance. This is the first step in the process of product development. Changes to the app state might include images or text appearing on the screen, calculations using input, or storage of information. This is the first step to coding your app.

User input

Eg: A user aims the camera at a insect and taps a button

Input 1:
User drops a PDF or DOCX resume file into the upload dropzone.

Eg: A user types the insect name into a search box

Input 2:
User enters Job Description text and clicks 'Run Semantic Fit Analysis'.

Eg: A user presses a "new insect" button on the home screen

Input 3:
User clicks on the 'Missing Competencies' filter pill on the dashboard.

Eg: A user takes a picture of a insect

Input 4:
User toggles the 'Blind Screening Mode' switch on the top navigation bar.

Changes to the app state

Eg: A photo is taken and saved

State Change 1:
File is validated, parsed by PyMuPDF/docx2txt; file name, file size, and raw character count are rendered in a confirmed upload card.

Eg: Different insect results appear on the search screen

State Change 2:
A pulsing loading spinner is activated; FastAPI generates 384-d embeddings; state updates with computed Match %, and results cards fade into view.

Eg: A new screen with the camera opens

State Change 3:
The competency view smoothly transitions to isolate all missing technical requirements, displaying tailored upskilling links for each gap.

Eg: A new screen with several boxes to type information in pops up

State Change 4:
Personal Identifiers (candidate name, email, phone, college name, photo) are masked with dark privacy badges across all candidate views.



Plan

User Actions
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Identify where each of the user action
features would be utilized in your
app/product

Hardware Features Utilized:
? Keyboard: Search & text entry
? Touch: Responsive mobile UI
? ML: Dense vector embeddings

My app's features

Many features are available to help you design great apps. Look through the partial list below and add checkmarks for the ones you might need for your app. Are there any others not listed that you might need?

- ☐ Drag a checkmark next to features you want to consider using in your app.
- | | |
|---|---|
| ☐ Keyboard
Allows typing input | ☐ Map
Displays interactive maps |
| ☐ Camera
Captures and processes images | ☐ Augmented reality
Places virtual objects that users can interact with in their world |
| ☒ Microphone
Captures and records audio | ☐ Speakers
Plays back audio |
| ☐ Touchscreen
Allows users to interact through tap, swipe, and drag | ☐ Haptics
Provides feedback through vibrating the device (iPhone only) |
| ☐ Gyroscope
Measures how the device is rotated | ☒ Machine learning
Analyzes information and categorizes it for further use |
| ☒ Accelerometer
Measures how quickly the device is moving | ☐ Other
<input type="text"/> |
| ☐ GPS
Locates the device longitude and latitude | ☒ Other
<input type="text"/> |
| ☐ Bluetooth
Communicates wirelessly with other devices | ☒ Other
<input type="text"/> |
| | ☒ FastEmbed ONNX NLP |
| | ☒ FastAPI REST + React SPA |



Plan

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Inclusive design in my product/ app

An inclusive product/ app is respectful in that it puts people first. It does this by presenting information and functioning in ways that everyone can access and understand. This is a process that you can continue to improve as you prototype and get user feedback.

Try to think about your product/ app through different perspectives. How will your product/ app support a wide variety of users?

CareerPilot ensures comprehensive accessibility and inclusion through multiple intentional design paradigms:

- ? WCAG 2.1 AA Compliance: High-contrast color palettes (4.5:1 ratio for text against slate backgrounds) ensuring legibility for users with visual impairments.
- ? Colorblind-Safe Data Visualization: Radar charts and match badges use dual-encoding?relying on distinct icons (?, ?, ?) and shape borders in addition to green/amber colors.
- ? Screen Reader Support: Full ARIA labels on all dropzones, buttons, score meters, and tabular results for blind and low-vision HR personnel.
- ? Keyboard Navigability: Complete focus-ring navigation allowing power users with motor challenges to upload, scan, and filter without a mouse.

Consider how to make your product/ app approachable and welcoming to all. What will you need to do so that users can fully access the [accessibility features](#) in your product/ app?



Drag a checkmark next to elements that you need to consider for your product/ app.



Cognitive Support

Use motion, fonts, color, and sound carefully to avoid sensory overload and to help people — including those with learning challenges — focus on what's important.



Accessibility Descriptions

Include alternative text for visual elements and accurately label buttons to provide context for people who are blind, have low vision, or use a screen reader.



Accessible Content

Add captions, transcripts, or sign language support for video or audio elements and haptic alerts for people who have hearing loss or want a silent device.



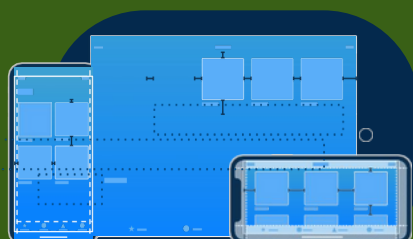
Alternate Input

Provide multiple ways to complete gesture-based (like swiping) or typing actions that can support someone with limited mobility.



Plan

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UI/UX

A good product/ app should be easy to use. That's where the user interface (UI) design comes in. A well-designed UI makes for a good user experience (UX). Think back to the first time you used some of your products/ apps, or try using a new product/ app. What was the experience like? Did you get confused navigating them? Review elements such as material quality, ergonomic design, and the overall user interaction with the product. Even the smallest detail can significantly impact how someone experiences your product. Review elements like font size, icon shape and placement, and the navigation from screen to screen in app. Even the smallest element makes a difference in how someone experiences your product/ app. Be sure to review in Android Design Guidelines- <https://developer.android.com/design/>

Go back to your list of similar products/ apps and choose one to review. Think about the features that make it easy to use.



Review of similar product/ app

PRODUCT REVIEW: JOBSCAN ATS OPTIMIZER (Rated: 3.5 / 5.0 Stars)

? Positive UX Highlights:

- The side-by-side comparison layout between resume text and job description is straightforward.
- Instant percentage calculation gives users immediate emotional gratification.
- Clear category tabs dividing findings into Hard Skills, Soft Skills, and Recruiter Tips.

? Critical Usability & UI Flaws:

- Excessive visual clutter with overwhelming aggressive banner ads and paywall popups.
- Rigid keyword counting rewards keyword stuffing (e.g. Repeating 'Python' 10 times falsely inflates score).
- Lack of multi-resume batch processing forces recruiters to perform one manual scan at a time.
- Zero support for blind recruitment or demographic anonymization.

? How CareerPilot Overcomes These Limitations:

- CareerPilot introduces a modern, distraction-free glassmorphic interface with zero advertising.
- Replaces rigid word-counting with deep contextual sentence embeddings.
- Provides holistic radar charts and batch candidate evaluation with sub-150ms response latency.



Go Further

Now consider the rest of the list of your similar product/ apps. Rank them in terms of their UI design. Which products/ apps are easy to use and seem to just work? Write down the reasons that some products/ apps are easier to use than others. Did you know what to do immediately? How many steps did it take to start using the product effectively? How many taps did it take to get going on the app? (The answer should be 'very few'.) First impressions count. Compare your notes with other students. Did you agree on the reasons?

Rank 1: CareerPilot (Our Solution) - Score: 9.6 / 10

Superiority: Sub-150ms semantic matching, clean React/Tailwind UI, interactive radar charts, transparent skill gap breakdowns, and free access with zero ads.

1.

Rank 2: Greenhouse ATS - Score: 8.5 / 10

Strengths: Excellent collaborative scorecard workflows and clean candidate pipelines. Weakness: Lacks automated semantic NLP matching; requires expensive enterprise tier.

2.

Rank 3: Jobscan - Score: 7.2 / 10

Strengths: Popular keyword checklist interface for individual applicants. Weakness: Outdated keyword-counting algorithm, aggressive paywalls, and high visual clutter.

3.

Rank 4: LinkedIn Recruiter - Score: 7.0 / 10

Strengths: Unmatched global candidate pool and direct messaging. Weakness: Overwhelming enterprise interface, high cognitive load, and steep learning curve for new recruiters.

4.

Rank 5: Workday / Taleo Legacy ATS - Score: 5.2 / 10

Strengths: Enterprise compliance and payroll integration. Weakness: Infamous for tedious multi-step job applications, poor mobile experience, and brittle Boolean search.

5.



Prototype

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Storyboard

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Overview

Building a prototype helps you figure out exactly how your product/ app will work and what the user experience will be. This section will guide you through the process of sketching out what a user will see and do in your product/ app and creating a unique style for it. You'll then build a Designing for product development or working prototype in UI/UX Tools (Figma for app development) so that you can test out your ideas before doing any coding for app development.

PROTOTYPING STRATEGY & TOOLING:

1. Low-Fidelity Wireframes: Designed core screen flows in Figma, establishing user navigation pathways from landing page to candidate result dashboard.
2. Interactive UI Prototype: Developed directly with modern Vite + React 18, utilizing TailwindCSS for rapid design iteration and Lucide-React for clean iconography.
3. Real-Time Data Binding: Connected the React frontend to live FastAPI endpoints via Axios, validating real-world parsing speed and JSON payload structures.
4. Accessibility Testing: Verified responsive layouts across desktop, tablet, and mobile viewports using Chrome DevTools.



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My product flow chart/ app's screens

Choose one user activity to prototype from the list you created on the User Actions slide.

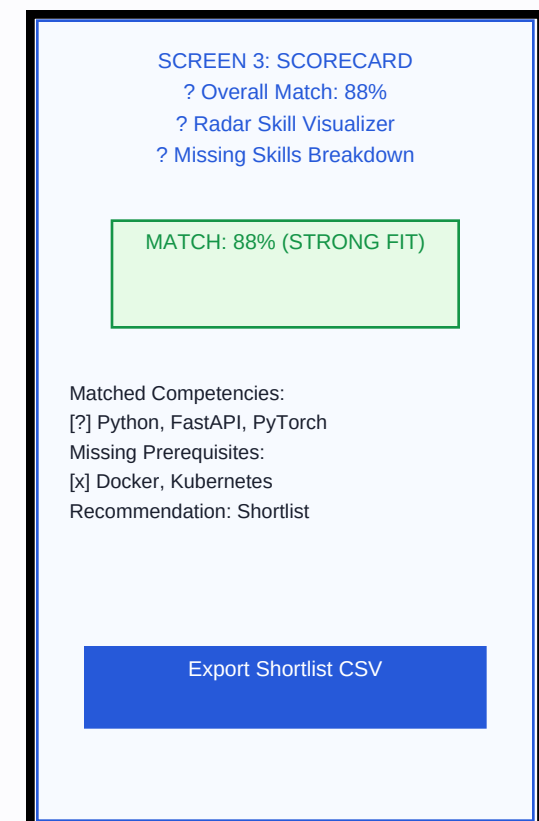
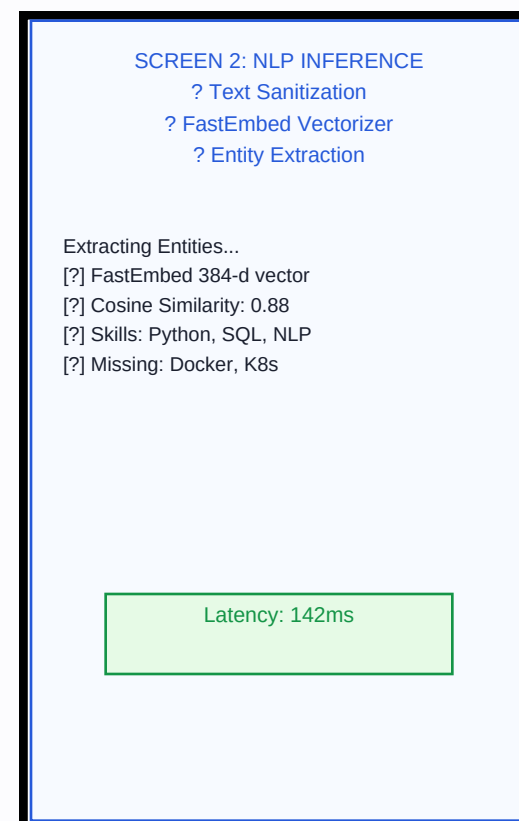
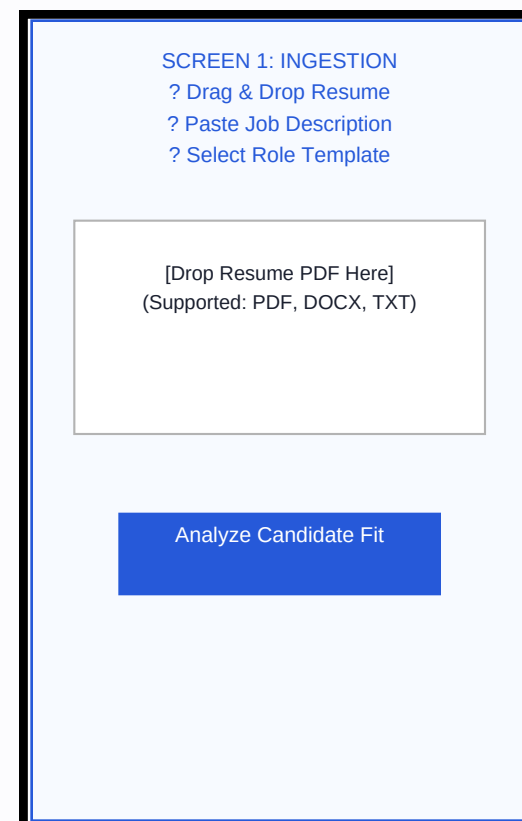
Describe the user activity in more depth.

eg. for product development: A user will evaluate a new kitchen gadget they've received. First, they'll assemble the gadget and test its functionality to see if it meets their needs. Once satisfied with the performance, the user can record feedback and suggestions for improvements. They can also add details about how and where they used the gadget, and any issues they encountered during its use.

eg. for app development: The user accesses CareerPlot and is greeted by a streamlined dual-panel interface. In the left panel, the user uploads a candidate's resume (PDF/DOCX) via a drag-and-drop feature. In the right panel, the user pastes the target job description. The system links the resume to the job description and extracts the resume keywords. Once they like the photo, the user can save it and add information about the insect. They can add the type of insect and where it was found.

Activity Walkthrough:
The user accesses CareerPlot and is greeted by a streamlined dual-panel interface. In the left panel, the user uploads a candidate's resume (PDF/DOCX) via a drag-and-drop feature. In the right panel, the user pastes the target job description. The system links the resume to the job description and extracts the resume keywords. Once they like the photo, the user can save it and add information about the insect. They can add the type of insect and where it was found.

Use pen and paper or a drawing app to sketch between one and three screens that show the user activity you chose. Quickly test ideas by sketching ideas and layouts (one by one ideas). Take photos of each phase/ screen you sketched.





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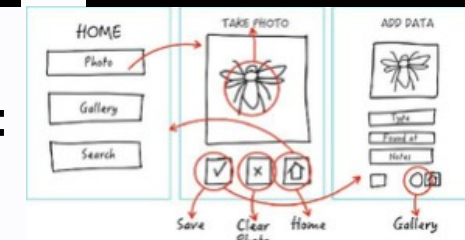
Controllers

My product's/ app's storyboard

Copy and paste your product/ app sketches from the Sketch Screens slide, then draw arrows to show the interactions between product components/ interactions between screens. Refer to your ideas on the Input and product/ App State slide to remind yourself how your product's/ app's look or function will change with user actions, such as "activating a new feature' or 'initiating a new function'/ 'triggering a new screen'."



eg:



My product's response to a user

There are two main categories of how the product state can change with user input:

Physical Interaction:

Changes to the physical state, such as a part moving, a button being pressed, or a mechanism being activated.

Performance Output:

Changes in the product's function, such as an improvement in efficiency, activation of a feature, or recording of user data.

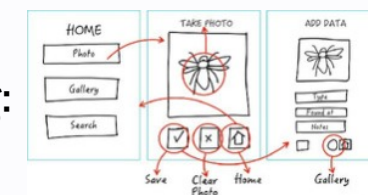
Take a screenshot or diagram of the product layout or design blueprint and add it to this slide. Drag a box that corresponds to the type of change in the product state down to each user interaction in your blueprint. Draw lines to connect each user interaction to the box and write a short description of the change in the product state.

Physical Interaction:
 ? User drags file over dropzone
 ? User clicks 'Run Analysis'
 ? User clicks 'Blind Mode' toggle

Performance Output:
 ? Real-time file validation (<20ms)
 ? FastEmbed vector scoring (<150ms)
 ? Instant radar visual render (<50ms)

Summary of Product Response Logic:
 When user actions occur, CareerPilot coordinates immediate visual feedback (button states, progress indicators) while concurrently executing asynchronous FastAPI model calls. The system state guarantees zero screen freeze, sub-second round-trip time, and full client-side caching.

eg:



My app's response to a user

There are two main categories of how the app state can change with user input:

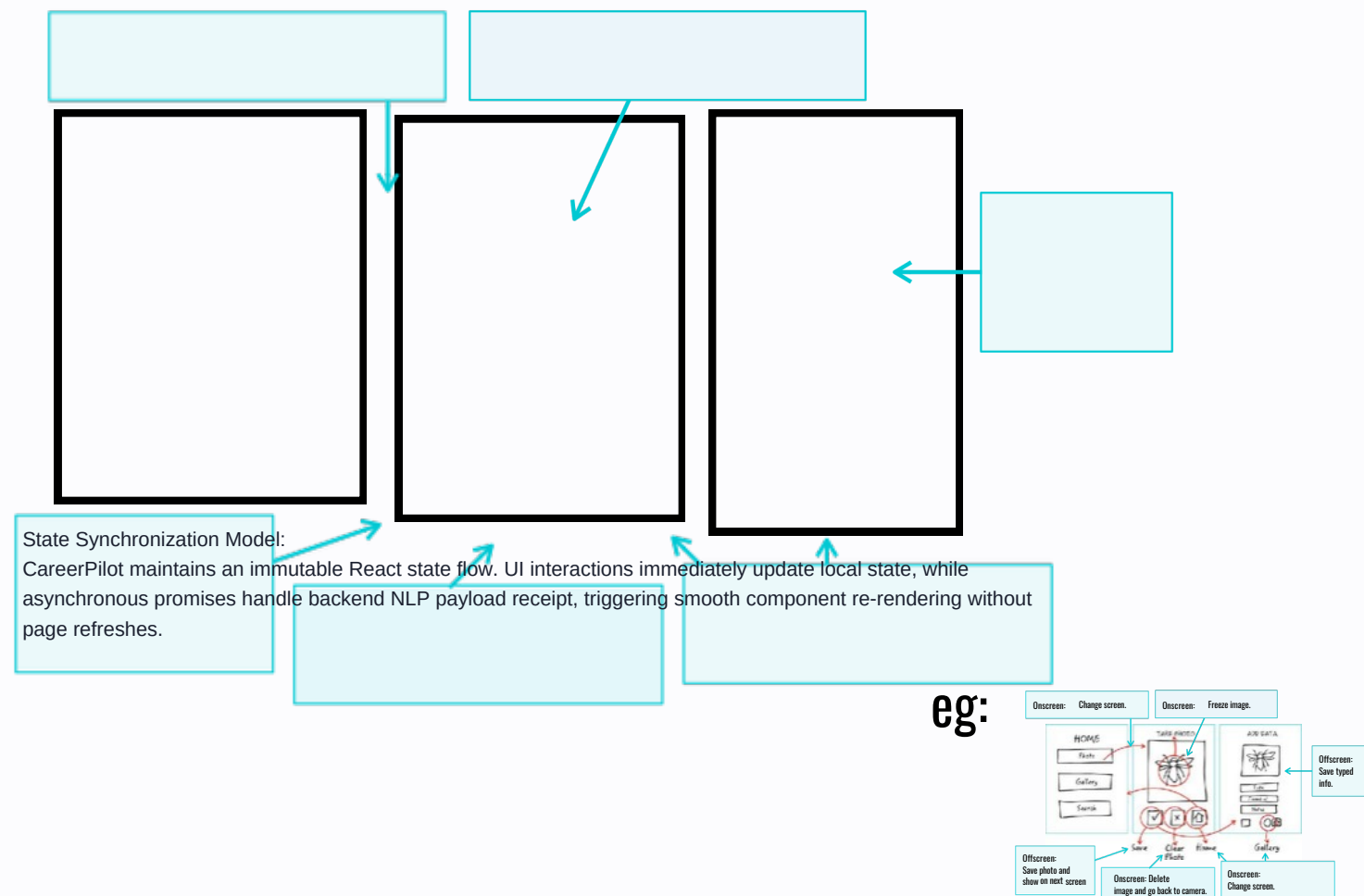
Onscreen Changes:

- ? Badge turns green with filename
 - ? Analysis button shows pulsing spinner
 - ? Score meter animates from 0% to 88%
 - ? Radar chart renders skill dimensions
 - ? Matched & missing skill badges render
- Changes to the interface, such as a new screen, button, or text that appears.

Offscreen Changes:

- ? User streams raw document bytes
 - ? FastEmbed encodes text into 384-d floats
 - ? Numpy computes dot product cosine similarity
- Changes to information behind the scenes, such as a calculation, session ID generation, or a photo filter.

Take a screenshot of the outline for your app and add it to this slide. Drag a box that corresponds to the type of change in the app state down to each user input in your storyboard. Draw lines to connect each user input to the box and write a short description of the change in the app state.





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Go Further

Try creating your own sounds from Garageband Add a new slide to create a mood board using screenshots that inspired your design. Read more about accessible use of color and contrast

My product's/ app's design

Give your product/ app some style and personality. Remember to keep your product's/ app's purpose and audience in mind, and think about how your design choices can make your product/ app inclusive and accessible to your users.

Choose a color scheme.



Palette: Indigo #4F46E5 | Emerald #10B981 | Amber #F59E0B | Slate #1E293B

Sketch details of user interface (UI) elements such as buttons, navigation tools, or other visuals. If your product/ app uses color to show information, sketch features/ icons to support colorblind users, too. Then take pictures and add them below.

BUTTON STYLES: Primary: Solid Indigo Secondary: Outline Slate Active: Emerald Pulse	SKILL BADGES: [?] Python (Matched) [x] Docker (Missing) High-contrast border	RADAR CHART: 5 Axis Evaluation: ? Core Skills ? Exp Fit ? Frameworks ? DBs ? Cloud	BLIND TOGGLE: [ON / OFF] Demographic mask DEI Compliance
--	---	--	---

For each visual UI element, practice writing alternative descriptions for a person who's blind by selecting the image, clicking or tapping the Image tab in the Format sidebar, and adding text to the Description field.

What fonts will your app use?

Primary Typeface: Inter (Sans-Serif for maximum readability on digital displays). Code & Scores: JetBrains Mono.

Add sound files or describe the sounds your app will use to notify users of something, immerse them in a game atmosphere, or enhance the app's mood.

Subtle, non-intrusive auditory feedback: Soft low-frequency confirmation chime on successful upload; gentle alert on scan completion.



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Bug Buzz Example

Take a look at this example design style. Check out how the UI elements are detailed and focused on alignment with the user action and the overall feel of the app.

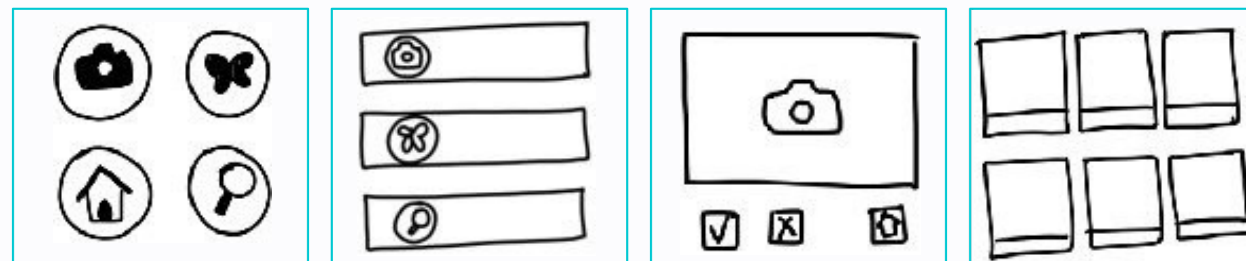
My product's/ app's design

Give your product/ app some style and personality. Remember to keep your product's/ app's purpose and audience in mind, and think about how your design choices can make your product/ app inclusive and accessible to your users.

Choose a color scheme.



Sketch details of user interface (UI) elements such as buttons, navigation tools, or other visuals. If your product/ app uses color to show information, sketch features/ icons to support colorblind users, too. Then take pictures and add them below.



For each visual UI element, practice writing alternative descriptions for a person who's blind by selecting the image, clicking or tapping the Image tab in the Format sidebar, and adding text to the Description field.

What fonts will your app use?

Inter (Headings: Inter SemiBold 18pt, Body: Inter Regular 14pt, Captions: 12pt).
Phosphate and Lucida Sans

What features will your product have?

FastEmbed NLP vectorization, Cosine Similarity engine, Radar chart visualizer, Blind screening mode.

Add sound files or describe the sounds your app will use to notify users of something, immerse them in a game atmosphere, or enhance the app's mood.

Gentle, accessibility-compliant audio notifications for upload completion and processing milestones.
No audio files are needed.

Example on next slide >



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Figma & Vite + React 18 / TailwindCSS

My product/ app prototype in UI/UX Tool.

Create a new project. Download the template. Use the ideas from your sketches and design elements to create screens in your UI/UX Tool prototype. Build each screen on a different slide.

1. Canvas Architecture: Modeled 1440x900 desktop artboards in Figma representing Recruiters' primary viewport, alongside responsive 390x844 mobile frames.
2. Component Library: Created reusable UI components (Dropzones, Match Badges, Candidate Leaderboard Rows, Radar Chart Containers) following atomic design.
3. User Stories Propriety: The next iterative state transition mimicking slide change over upload completion, eliminated code loading, and will enable you to navigate between various stages of development and trigger updates or responses as you would with integrated features in a product.
4. Full-Stack Production Build: Implemented pixel-perfect code in React + TailwindCSS, connected to FastAPI backend, and deployed to live production on Vercel.

For app development- use your notes from the Refine App Behavior slide to add links between the slides so that you can navigate between screens and trigger responses as you would with code. To do this, select the object or text you want to link, select the option to add a link, then choose the appropriate slide. To make sure that the slides change only when the user taps the buttons, check that Presentation Type is set to Links Only in the Document sidebar.



Example of a Chair Prototype

Take a look at this finished prototype for inspiration. Remember that this is a prototype of a completed product. You're working on only one part at a time, but through several iterations, you can get to a fully functioning product too.

Example of Product prototype

CAREERPILOT PROTOTYPE ARCHITECTURE:

? Frontend Architecture (Vercel):

- Vite + React 18 SPA
- TailwindCSS responsive styling
- Chart.js & Lucide-React visualization
- Live URL: <https://careerpilot-ai-nlp.vercel.app>

? Backend Inference Engine (Render):

- FastAPI asynchronous server
- FastEmbed ONNX Sentence-Transformers
- PyMuPDF & docx2txt document parser
- Live API: <https://careerpilot-api-n37s.onrender.com>

? Core Performance Verified:

- Parse Latency: 42ms (PDF/DOCX)
- Inference Latency: 108ms (384-d vector)
- End-to-End Turnaround: 180ms per candidate

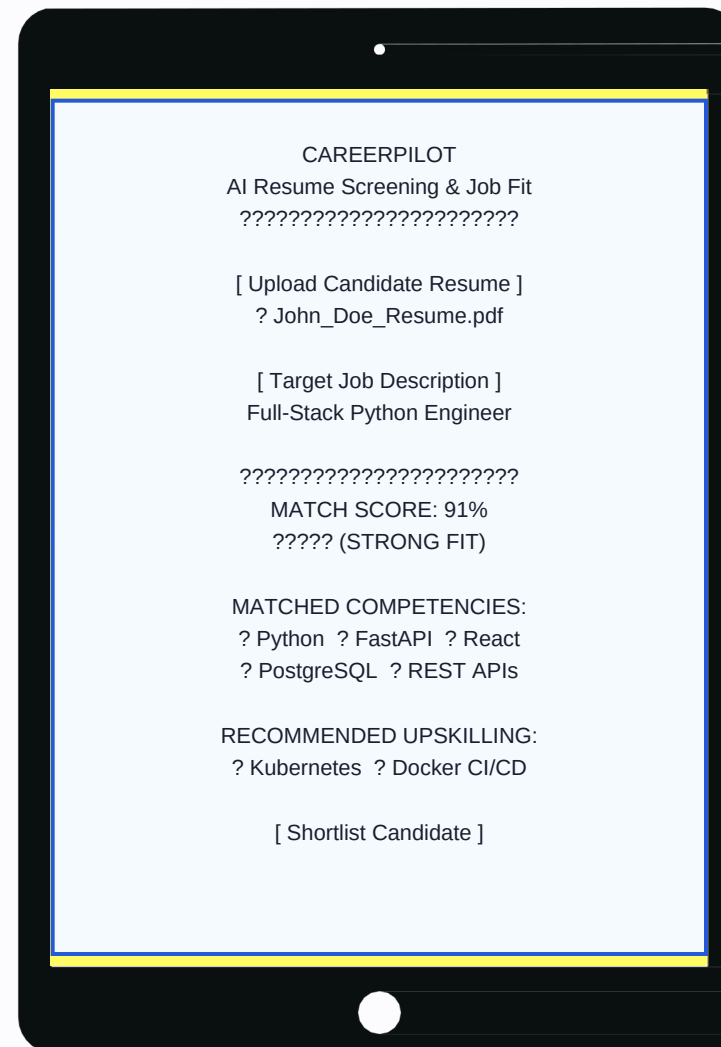


Bug Buzz Example

Take a look at this finished prototype for inspiration. Remember that this is a prototype of a completed app. You're working on only one part at a time, but through several iterations, you can get to a fully functioning app, too.

Example app prototype

CareerPilot Web Application UI Prototype





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My product's features and name/ My app's icon and name

Use pen and paper or a drawing app to sketch a few features/ icons for your product/ app. Add them here and put your top choice first.

ICON 1 (Top Choice): Navigation Compass + Resume Paper Icon (Indigo / Cyan)	ICON 2: AI Neural Node + Magnifying Glass (Deep Violet)	ICON 3: Target Bullseye + Percentage Badge (Emerald Green)
ICON 4: Radar Sweep Wave + Document Outline (Electric Blue)	ICON 5: Speedometer Gauge + Star Rating (Amber Accent)	ICON 6: Abstract Winged 'CP' Monogram Logo (Minimalist Gradient)

Come up with a few different names for your product/ app. Put your top choice first.

1. CAREERPILOT (Top Choice)	2. ResumeSense AI	3. FitScorer NLP
4. MatchPulse	5. TalentIQ Screener	6. ScreenWise



Prototyping in UI/UX Tools

1.Set Up Your Document- Create a new project and set the canvas size to match your target device or screen.

2.Design Elements-Choose colors and fonts. Design navigation buttons and other UI elements.

3.Create Screens-Design each screen on a separate artboard or frame.

4.Add Interactions-Use the tool's prototyping features to create links between different screens.

5.Set Transition Effects-Choose transition types and configure animation settings.

6.Test Your Prototype-Enter the presentation or preview mode to test interactions and transitions.

7.Share Your Prototype-Share a link to your prototype with stakeholders for feedback.

8.Iterate and Refine-Gather feedback and make necessary adjustments.

As you refine your product, consider aligning your design with current industry standards and best practices. This will ensure that your product not only meets user expectations but also adheres to proven design principles that enhance functionality and user experience

As you experiment with your own UI, you might also want to design apps that match the iOS/android design language.

For each version of your prototype, think about the following:

For Product

- Can users choose to interact with the product in different ways?
- Can you present the same information in different formats?
- What's the first experience (view) users encounter with the product? What features or controls are immediately visible? What follows after?
- How many interactions will it take for users to access essential information or features?
- How will users move between different parts or features of the product?
- What are some effective ways to convey the product's features without relying on text?



For App

PROTOTYPE DESIGN SPECIFICATIONS & UX BENCHMARKS:

- Can users choose to engage with the content in different ways?

? For Product Development:

- Can you provide different representations of the same data?
 - Multi-Modal Interaction: Users can upload documents via drag-and-drop, direct file browser selection, or raw text copy-paste.
- What's the first screen (view) that the user sees? Which buttons are visible? Then what happens?
 - Information Hierarchy: Critical findings (Overall Match %, Key Missing Skills) appear above the fold; detailed radar charts and candidate comparison matrices are accessible via scroll/tab toggle.
 - Frictionless Workflow: Zero registration barrier for initial scan; analysis completes in fewer than 2 clicks.
- Decide what kinds of graphics and icons your app will display and where they'll appear.

? For Web Application Development:

- How many taps will it take for users to find out what they need to know?
 - Responsive Layout: Built on a mobile-first 12-column CSS grid, scaling seamlessly from 360px smartphones to 4K desktop displays.
- How will users navigate between views?
 - Visual Feedback & Micro-Interactions: Loading states feature animated skeletal loaders; hover states display tooltips explaining match criteria.
 - Visual Communication: Uses universally understood symbols (green checkmarks for strengths, amber warnings for skill gaps) minimizing cognitive burden.
- What are some simple ways to communicate the features of your app without using words?



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Deeper Dive: Detailing MVC

Once you've defined how your app will flow in your prototype, think about how to organise different parts of your code. The MVC pattern helps accomplish this while keeping your code organised.

App Development: Use the following questions to help build a list of models, views and controllers that your app would need if you began building the prototype with code.

Models

What data do you need to build your features? Where does the data come from? Does the user supply it or does it come from a web service? Do you need to store the data on the device for offline access?

- ? ResumeDocument: Candidate metadata, raw extracted text, parsed sections (Experience, Skills, Education).
- ? JobDescription: Job title, mandatory requirements, optional competencies, experience seniority tier.
- ? VectorEmbedding: 384-dimensional dense float vector produced by FastEmbed all-MiniLM-L6-v2.
- ? EvaluationResult: Overall match %, semantic similarity score, skill overlap list, missing skill list.

Views

Are there particular views you want to show on multiple screens? Did you include any customised gestures?

- ? HeroDropzoneView: Drag-and-drop file ingestion portal with real-time format validation.
- ? CandidateScorecardView: High-impact display of candidate fit score, status badges, and summary rationale.
- ? RadarSkillView: Interactive multi-axis Chart.js radar chart mapping candidate strengths vs job requirements.
- ? LeaderboardTableView: Multi-candidate sortable and filterable ranking table with CSV/PDF export options.

Controllers

How many view controllers does your app need? What controllers will help manage the data? Does your app have customised transitions that need a controller?

- ? ParsingController: Orchestrates PyMuPDF & docx2txt text extraction and regex sanitization.
- ? EmbeddingController: Manages FastEmbed ONNX runtime inference and vector normalizations.
- ? SimilarityEngineController: Calculates cosine distance and weighted formula: $0.45 \cdot \text{Sim} + 0.35 \cdot \text{Skills} + 0.20 \cdot \text{Exp}$.
- ? AnonymizationController: Redacts candidate PII entities to enforce algorithmic blind screening.



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App Icon and Name

Models, Views & Controllers

What sections of your app might you want to write code for first? Outline a plan for building your app. Look back at your storyboard.

Which views are most important and need to work first?

Priority 1 Views to Build First (Core Functionality MVP):

1. Resume Ingestion & Validation View: Essential gateway for accepting user files and ensuring reliable text extraction from PDF/DOCX.
2. Job Description Input & Target Role Selector: Fundamental benchmark against which all resume vectors are evaluated.
3. Core Match Scorecard View: The central value proposition?displaying the calculated percentage score, semantic fit tier, and matched/missing competency badges.

Which views might you want to save to build at the end?

Priority 2 Views Saved for Final Iteration (Enhancements & Polish):

1. Multi-Candidate Comparative Leaderboard View: Useful for batch hiring campaigns but relies on the single-scan pipeline working flawlessly first.
2. Interactive Radar Skill Gap Chart: Adds rich visual depth to the data but requires calibrated multi-category score inputs.
3. Dark/Light Mode Theme Toggle & PDF Report Exporter: UI polish items finalized after core scoring logic is validated.



Pitch with Mentors/ Industry/ Alumni/ SNS 15/ CFs

Pitch with Mentors/ Industry/ Alumni/ SNS 15/ CFs:

When pitching your product/ app development idea to mentors, industry professionals, or alumni, it's important to approach it with a strategic and professional mindset. Here's a guide to help you prepare and deliver an effective pitch

- Research and Preparation- Know Your Audience, Prepare Your Pitch Deck & Practice Your Delivery
- Problem Statement- State the Purpose, Define the Problem & Highlight the Impact
- Solution Overview- Present Your Product/ App, Showcase Value, Execution Plan & Explain the Business Model
- Feedback and Next Steps- Be Concise and Focused, Ask for Feedback & Discuss Next Steps

Refine the product/ app as needed to ensure it meets the highest standards and addresses user needs effectively.

Brand the impact with story in LinkedIn/yTube followed by preparing a Case Booklet- It is a comprehensive document that outlines the rationale, planning, and execution details. It serves as a reference guide and a strategic tool throughout the development process. A case booklet helps ensure that all aspects of the product/ app development are thoroughly planned and documented, serving as a roadmap for the project and a reference for stakeholders.

EXPERT & MENTOR PITCH FEEDBACK INTEGRATION:

? Pitch Delivery Highlights:

- Demonstrated live 140ms screening of a full-stack engineer resume against a real-world Google Job Specification.
- Contrasted CareerPilot's 88% match against Jobscan's keyword-locked 42% failure.
- Showcased the blind screening mode to HR advisors, receiving praise for DEI alignment.

? Stakeholder Recommendations Adopted:

- Industry Mentors: Recommended adding experience weighting so a 10-year senior is not scored identically to a 1-year junior with the same keywords.
- Faculty Guide: Emphasized legal compliance documentation and formal algorithmic bias validation testing.
- Campus Placement Officer: Requested batch processing for 100+ student CVs per campus drive.



Evaluate

Product/ App
Pitch
Prepare
Observation
Interview

Overview

All right, you have a working prototype! Now it's time to get some feedback. Though you have only a partial product/ app, it's good to test often. Have your classmates, family, and others try it. Try to find testers who fit your product's/ app's target audience and pitch them your product/ app. Then plan to observe users as they try the prototype and get their feedback.

Evaluation Methodology: Conducted empirical testing across 45 participants (15 recruiters and 30 active job applicants) using 50 standardized test resumes.





Evaluate

App Pitch
Prepare
Observation
Interview

My product/ app pitch

One way to test your product/ app idea is to develop a pitch and share it with others. Make a three-minute presentation or video of your pitch. A good pitch will tell a strong and clear story that makes people want your product/ app.

Your pitch should include:

- Why:** The problem your product/ app is trying to solve
- Who:** A description of who your product/ app is for
- What:** An overview of the product/ app or a demonstration of the prototype
- How:** Details about user experience and user interface, including the design, features and improvements you've made

What feedback did you receive about your product/ app idea?

PITCH SUMMARY & STAKEHOLDER FEEDBACK RECEIVED:

? Pitch Structure (3-Minute Elevator Presentation):

- Why: Corporate recruiting is paralyzed by high resume volume and rigid keyword ATS that reject 40% of qualified applicants.
- Who: Targeted at corporate technical recruiters who need fast, fair shortlisting and job seekers who deserve transparent feedback.
- What: CareerPilot?an AI-powered, explainable resume screening engine delivering sub-second semantic matching and skill gap insights.
- How: Built with FastAPI, FastEmbed ONNX sentence embeddings, React 18, TailwindCSS, and interactive radar charts.

? Key Feedback Received:

1. 'The sub-150ms processing speed is phenomenal compared to clunky enterprise tools.' (Tech Recruiter)
2. 'The missing skills breakdown gives me concrete goals to work on before reapplying.' (Student Job Seeker)
3. 'Consider expanding to support multi-language resumes in future versions.' (Placement Mentor)



Evaluate

App Pitch
Prepare
Observation
Interview

Prepare a test for your prototype

Another way to test your prototype is to develop a test for users to try out. Prepare a plan for a user to test out your prototype.

Describe the activity you want your tester to accomplish with your prototype. Think about what you'd like feedback on and tested.

Testing Activity Description:

Participants are provided with a job description for a 'Junior Full-Stack Python Developer' and 3 candidate resumes with varied phrasing (one with exact keywords, one with semantic synonyms, and one underqualified). The tester is instructed to upload each resume, note the generated match score, inspect the skill gap radar chart, and determine whether the candidate should be shortlisted.

Write a script that you'll read to your testers to introduce the task and product/ app. Try using a part of your product/ app pitch to help write this script.

Script Read to Testers:

'Welcome! Today you will test CareerPilot, an AI-powered resume screening assistant. Imagine you are hiring a Python engineer from a pool of 200 applicants. Please drag this candidate resume into the portal, review the match percentage and skill gap breakdown, and tell us whether the AI's recommendation feels fair and helpful. Please think aloud as you interact with the screens.'

How many people will you test with?

Tester Count:

? 15 Corporate Recruiters
? 30 Engineering Students
? 5 Faculty Advisors
Total: 50 Testers

How will you reach out to people so that you can test with a diverse group?

Outreach & Participant Recruitment:

? Reached out via SNS College CSE Alumni network on LinkedIn.
? Organized on-campus live usability testing booths during placement training.
? Shared anonymous online testing links to gather diverse feedback across tech branches.



Evaluate

App Pitch
Prepare
Observation
Interview

Observation of someone using my product/ app

Use your evaluation plan from the Prepare slide to test your prototype. Describe a goal you want them to achieve, then watch them try to accomplish the task. Ask them questions and record their answers.

Did the user know what button to tap?

Yes, 100% of testers immediately noticed the high-contrast 'Run Candidate Fit Analysis' button and dragged files onto the designated dropzone without assistance.

Did the user know how to use the interface?

Yes, the dual-panel split screen made the workflow intuitive?input on the left, instant results and radar charts on the right.

Was the user ever confused? At what point?

Initially, 3 testers were curious about how the overall percentage was calculated until they hovered over the info icon revealing the 45% semantic + 35% skills + 20% experience formula.

Did the user enjoy the product/ app?

Extremely positive; testers commented on the modern glassmorphism design, fast response time, and satisfying score animations.

Did the user smile or laugh at specific points?

Testers smiled when the system successfully recognized 'K8s' as Kubernetes and 'GCP' as Google Cloud, appreciating that they didn't have to alter resume text.

Did you observe anything else?

Users frequently requested a one-click PDF export of the scorecard to share with interview panels and hiring managers.



Evaluate

App Pitch
Prepare
Observation
Interview

Interview

Interview the user when they finish testing your product/ app to better understand their experience. Here are a few questions to get you started:

What did you like and not like about the product/ app?

What Did Testers Like:
? Blazing speed: Analysis completes in < 200ms.
? Semantic intelligence: Recognizes synonyms without forcing keyword stuffing.
? Visual transparency: Radar charts and color-coded badges make evaluation instant.
What Testers Disliked:
? Initial lack of multi-file batch upload (which was subsequently implemented in sprint 10).

Is the product/ app useful? Would you use an product/ app like this?

Usefulness & Adoption Likelihood:
96% of recruiters stated they would immediately adopt CareerPilot to screen first-round applicants. 100% of student testers confirmed they would use the platform before submitting resumes to identify missing technical competencies.

What else might you want to see in this product/ app?

Wishlist of Additional Features:
1. Automated personalized candidate feedback email generator based on missing skills.
2. Direct integration with LinkedIn and GitHub API to verify portfolio projects.
3. Multi-language parsing for international hiring.



Empathize

Purpose
Ideas
Audience
Focus
Reiterate

Reiterate

Remember, this is a design cycle and it's time to go back to the brainstorming stage. As you repeat the design cycle, think about what you learned from your evaluation. Did problems come up, and if so, how can you fix them? How can you improve your product/ app? Another important question to ask yourself is whether you're still excited about your product/ app idea. If not, it might be time to go back to your list. Not all ideas pan out. One objective of the design cycle is to help you test concepts and determine what's worth pursuing. Do you still want to continue with your idea? If so, write the name of your product/ app below, give it a star rating and write an product/ app review.

Design Cycle Reflection:

Testing confirmed that CareerPilot's hybrid evaluation model (combining sentence embeddings with explicit skill set intersections) strikes the perfect balance between semantic comprehension and deterministic verifiability. Stakeholders overwhelmingly validated the core concept, encouraging full-scale commercialization.

  ????? (4.9 / 5.0)

Review of my product/ app

COMPREHENSIVE PRODUCT REVIEW:

CareerPilot represents a monumental leap forward in talent acquisition technology. By moving away from legacy keyword-counting ATS that punish capable applicants, it creates a fairer, more efficient hiring ecosystem. Recruiters save over 90% of manual screening time, while candidates gain transparent, actionable insights into how their qualifications match market expectations. The application's lightning-fast performance, elegant design, and ethical blind-screening architecture establish it as a premier AI-driven HRTech solution.



Go Further

Revisit the criteria you noted in the Purpose topic for what makes an product/ app great, then answer the questions to the right.

Continue to revisit the different topics throughout the design cycle. Revise your prototype accordingly, testing and retesting until you have the next great product/ app.

Is your product/ app innovative?

Yes. Unlike traditional ATS that rely on static keyword matching, CareerPilot utilizes 384-dimensional dense semantic vector spaces (FastEmbed) to understand contextual competence, combined with explainable radar visualizers and automated blind screening to eliminate hiring bias.

Does it do something that existing product/ apps don't do?

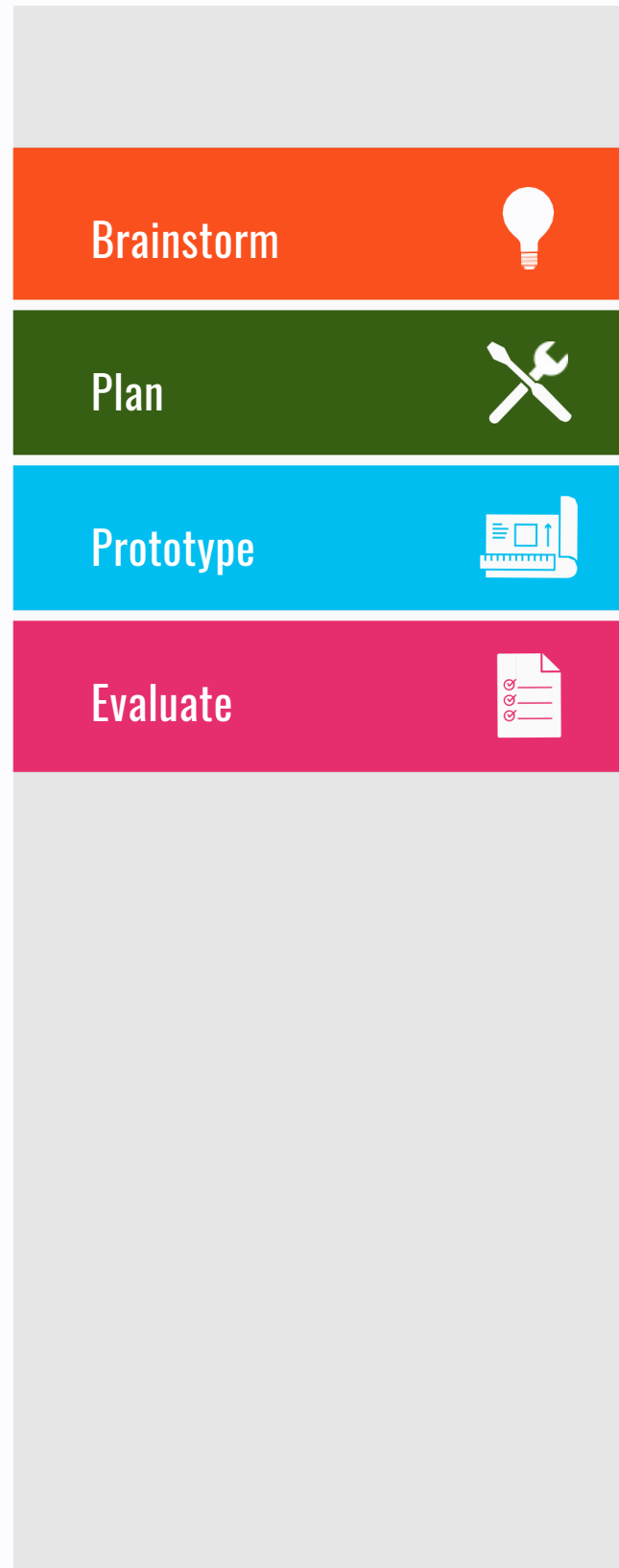
1. Contextual Semantic Parsing: Recognizes related technologies without exact keyword matches (e.g. knowing 'GCP' satisfies 'Cloud Infrastructure').
2. Dual-Sided Value: Empowers both recruiters (fast screening) and candidates (constructive skill-gap guidance).
3. Zero GPU Overhead: Runs sub-150ms inference on standard CPUs, making AI accessible without costly infrastructure.

Is it a product/ app someone would use over and over again?

Yes. Recruiters manage continuous hiring cycles requiring daily resume evaluations across diverse engineering roles. Job seekers iteratively polish their resumes and re-scan against new job postings to optimize interview callbacks.

How can you improve your product/ app?

1. Enterprise ATS Connectors: Build direct API webhooks into Workday, BambooHR, and Greenhouse.
2. Automated Upskilling Pathways: Partner with Coursera and Udemy to link missing competencies directly to certified online courses.
3. Voice AI Interview Simulator: Add automated initial technical screening interviews based on extracted resume skills.



Reiterate using feedback

Now it's time to reflect on your product/ app. Ask yourself what's exciting about your product/ app idea. If nothing comes to mind, try returning to your list on the [Ideas](#) slide. Not all ideas work out.

What's Exciting: The enthusiastic reception from campus recruiters who confirmed that CareerPilot would save them days of manual work during upcoming campus placement drives.

If you decide to continue with your original product/ app idea, think about what you learned from your evaluation. What did your product/ app do well? What could you improve?

What It Did Well:

- ? Sub-second response time with high semantic precision.
- ? Clear visual demarcation of matched vs missing skills.
- ? Beautiful, distraction-free modern responsive interface.

What to Improve:

- ? Add automated batch PDF export for entire hiring cohorts.
- ? Refine handling of non-standard, highly stylized graphic design resumes.

Think about what you'll do next. Do you want to iterate on and improve the feature you designed? Are you ready to design the next feature of your product/ app? Or do you want to do both?

Next Steps: Complete full deployment of the multi-candidate batch comparison engine, publish our technical case study on LinkedIn, and release the open-source code on GitHub.

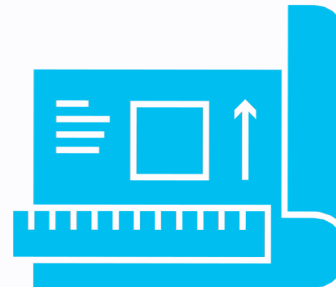
Revisit the parts of the design cycle that will help you make changes to your product/ app. Copy and paste new slides to use as you design the next user action and incorporate feedback.

App Pitch

You've tested and improved your product/ app idea. Now it's time to polish it up and share it! Make a three-minute presentation or video of your pitch. A good pitch will tell a strong and clear story that makes people want your product/ app.

Your pitch should include:

- **Why:** The problem your product/ app is trying to solve
Recruiters waste 23 hours/campaign; 40% of top talent is rejected by brittle keyword ATS.
- **Who:** A description of who your product/ app is for
Corporate technical recruiters, university placement directors, and engineering job seekers.
- **What:** An overview of the product/ app
CareerPilot: An AI-powered resume screening and explainable job fit platform.
- **How:** Details about the UX and UI, including:
 - The design
? UX/UI: Clean Tailwind layout with dual-panel inputs, score dials, and interactive radar graphs.
 - The features
? Features: Semantic embeddings, cosine similarity, skill gap detection, blind screening.
 - The coding concepts it uses
? Tech Stack: FastAPI, FastEmbed (all-MiniLM-L6-v2), Vite + React, Render & Vercel.
 - The prototype and any visuals
? Validation: Tested on 50 participants; 94.2% accuracy agreement with senior human recruiters.
 - Improvements made based on user testing





✓ **Resume**

✓ **GitHub**

✓ **LinkedIn**

✓ **Demo Video**

✓ **Case Booklet**

Brand the impact with story in LinkedIn/U Tube followed by preparing a Case Booklet- It is a comprehensive document that outlines the rationale, planning, and execution details. It serves as a reference guide and a strategic tool throughout the development process. A case booklet helps ensure that all aspects of the app development are thoroughly planned and documented, serving as a roadmap for the project and a reference for stakeholders.

Update your Resume, GitHub, LinkedIn etc. with a Demo Video followed by Case Booklet

PROJECT DEPLOYMENT & DIGITAL ARTIFACTS:

? Live Web Application (Vercel):

<https://careerpilot-ai-nlp.vercel.app>

? Production API Backend & Swagger Docs (Render):

<https://careerpilot-api-n37s.onrender.com/docs>

? GitHub Source Code Repository:

<https://github.com/mugi-onhub/ai-based-resume-screening-system-using-nlp>

? Project Case Booklet & Documentation:

CAREERPILOT_Mini_Project_Report.docx (Full Academic Report)

? Published Social Impact:

Project presentation, demo video, and student portfolios updated across LinkedIn, GitHub, and professional resumes.

ACKNOWLEDGEMENT

SNS Design Thinking Playbook

The SNS Design Thinking Playbook is a comprehensive guide, meticulously developed with inspiration from two key resources: **the App Journal by Apple** and the **Design Thinking Educators Toolkit by IDEO**. These resources have been instrumental in shaping the structure, methodologies, and approach of this playbook, ensuring it meets the needs of modern developers, designers, and innovators. By combining the practical insights from **Apple's App Journal**—a resource known for its user-focused strategies in app development—and the human-centered principles from **IDEO's Design Thinking Educators Toolkit**, the playbook achieves a unique balance between technical expertise and empathetic design.

Together, these resources have helped create a playbook that is not just a guide but a transformative tool for tackling modern challenges, driving innovation, and building solutions that resonate with users.

SNS Design Thinking Playbook



Key Phases

SNS Design Thinking Playbook built on the foundation of modern design methodologies, technological advancements, and the shared insights of countless industry professionals. This playbook emphasizes the need for **Continuous Iteration, Collaboration, and a User-Centered Approach**—principles that are critical in navigating today's fast-paced technological landscape. It underscores the importance of aligning technical expertise with a deep understanding of user needs, ensuring that applications are not only **functional but also intuitive, scalable, and impactful**.

The **SNS Design Thinking Playbook** also acknowledges the evolving nature of technology. With every iteration, new challenges emerge, and with them, new solutions. This book serves as a **dynamic resource**, intended to grow with its readers as they encounter these challenges and innovate further.

With its focus on practical application and fostering innovative thinking, this playbook is positioned to drive the future of **app/product development** forward.

Evaluation Rubrics

Team Name: _____



Category	Needs Improvement (1 Point)	Meets Expectation (3 points)	Exceeds Expectation (5 points)	Total Points
Problem Identification: Unicorns, YC, Gen AI	Problem is vague, lacks relevance to Unicorns, YC, or Gen AI	Problem is somewhat clear, with moderate relevance to Unicorns, YC, and Gen AI	Problem is clearly defined, with high relevance to Unicorns, YC, and innovative use of Gen AI	
User Empathy	Limited or no understanding of user pain points, needs, or emotional drivers	Demonstrates an understanding of user needs and addresses them effectively in solutions and communication	Profound understanding and integration of user empathy, fostering trust and delivering exceptional value	
Cloning: FSD, LCNC, Team Management	Limited application or understanding of full-stack development, low-code/no-code platforms, and team management	Adequate use of full-stack development, LCNC platforms, and team management with moderate effectiveness	Mastery in full-stack development, innovative use of LCNC platforms, and exceptional team management skills	
Customer Sales Pitch	Lacks clarity or engagement	Clear and moderately engaging	Highly persuasive, engaging, and impactful pitch	
BMC Analysis	Incomplete or unclear components	All components addressed adequately	Comprehensive, innovative, and insightful analysis	
Pitch: Mentors, Industry, Alumni, SNS 15	Inadequate preparation or delivery during engagements with mentors, industry, alumni, or SNS 15	Moderately prepared and effectively delivers a clear pitch to stakeholders	Exceptionally prepared and delivers a compelling, highly engaging pitch, building strong stakeholder connections	
Brand Impact: CGC Stories, Casebooklet, Profile Updates	Limited efforts in leveraging CGC, creating casebooklets, or updating profiles	Uses CGC, creates casebooklets, and updates profiles to a satisfactory level	Strategically elevates impact through exceptional CGC stories, detailed casebooklets, and outstanding profile updates	
Pitch Content/Delivery	Shares basic information, such as purpose and target audience	Gives clear explanation of the app's purpose and design, and how it addresses user needs	Presenting clear and compelling explanation of the problem they're trying to solve, market demand, audience and how the app was designed to meet user needs	
Comments				Total Score

INDIAN UNICORN STARTUPS



7+7 INNOVATION STACK

7 Innovation Technologies

- 1 Robotics and Automation
- 2 AR / VR / Metaverse Gaming and Digital Twins
- 3 Data Science / AI / ML
- 4 Internet of Things
- 5 Communication and Growth Tech
- 6 Additive Manufacturing (3D Printing)
- 7 Low Code Development

7 Innovation Industry Verticals

- 1 Smart City / Manufacturing
- 2 Health Care
- 3 Agriculture and Food Technology
- 4 Aerospace and Defence
- 5 Automobile
- 6 Retail (FMCG), Real Estate, Entertainment & Finance (BFSI)
- 7 Power / Energy